Implicit Differentiation and Inverse Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumDuring Section II Part A (30 minutes, Calculator Active), a student completes Questions 1 and 2 with 5 minutes remaining. When time is called and calculators are stored for Part B (60 minutes, No Calculator), may the student return to Questions 1 and 2 to add work, and what restrictions apply?

Yes. Students are permitted to read, edit, and add work to Questions 1 and 2 at any point during the remaining 60 minutes of Section II Part B. However, they may NOT use a calculator under any circumstances during Part B. Any remaining work on Questions 1 and 2 must be written using analytic notation, non-calculator algebra, or conceptual justifications.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

[Vectors & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Unit 9 · Parametric Equations & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

5 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[AP Calculus BC Exam Format & Units Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Functions (HL Extensions) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

Compute derivatives: $\mathbf{r}'(t) = \langle 1, 2t, 2t^2 \rangle$ $\mathbf{r}''(t) = \langle 0, 2, 4t \rangle$ At $t = 1$: $\mathbf{r}'(1) = \langle 1, 2, 2 \rangle \Rightarrow |\mathbf{r}'(1)| = \sqrt{1 + 4 + 4} = 3$ $\mathbf{r}''(1) = \langle 0, 2, 4 \rangle$ Tangential acceleration a_T: $a_T = \frac{\mathbf{r}' \cdot \mathbf{r}''}{|\mathbf{r}'|} = \frac{(1)(0) + (2)(2) + (2)(4)}{3} = \frac{0 + 4 + 8}{3} = \frac{12}{3} = 4$ Normal acceleration a_N: Compute cross product at $t = 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

1. Step formula: $x_{n+1} = x_n + h$, $y_{n+1} = y_n + h(x_n + y_n)$ with $h = 0.1$, $x_0 = 0$, $y_0 = 1$. Step 1 (to $x_1 = 0.1$): $y_1 = y_0 + 0.1(x_0 + y_0) = 1 + 0.1(0 + 1) = 1 + 0.1 = 1.1$. Step 2 (to $x_2 = 0.2$): $y_2 = y_1 + 0.1(x_1 + y_1) = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.1(1.2) = 1.1 + 0.12 = 1.22$. Euler's approximation is $y(0.2) \approx 1.22$. 2. Exact analytical solution: Rewrite as $\frac{dy}{dx} - y = x$. Integrating factor $\mu = e^{-x}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · Advanced / AP FRQOn an AP Calculus BC exam question asking whether a series $\sum_{n=1}^{\infty} a_n$ converges, a student writes: "Since $\lim_{n \to \infty} a_n = 0$, the series converges." Evaluate the mathematical validity of this statement, identify the logical fallacy, and provide a standard counterexample that disproves it.

Evaluation: The statement is mathematically invalid and earns 0 points. Logical Fallacy: The student has committed the fallacy of the converse. The n-th Term Test states: If a series $\sum a_n$ converges, then $\lim_{n \to \infty} a_n = 0$. The contrapositive is also true: If $\lim_{n \to \infty} a_n eq 0$, then the series diverges. However, $\lim_{n \to \infty} a_n = 0$ is a necessary condition, NOT a sufficient condition for convergence. Counterexample: The harmonic series.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Let $\mathbf{a} = (2;\ -1; 3)$ and $\mathbf{b} = (1; 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

1. Compute the cross product via determinant: $\mathbf{a} \times \mathbf{b} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 2,\ -1, 3; 1, 4,\ -2| = \mathbf{i}[(-1)(-2) - 3(4)] - \mathbf{j}[2(-2) - 3(1)] + \mathbf{k}[2(4) - (-1)(1)]$ $= \mathbf{i}(2 - 12) - \mathbf{j}(-4 - 3) + \mathbf{k}(8 + 1) = -10\mathbf{i} + 7\mathbf{j} + 9\mathbf{k} = (-10; 7; 9).$ 2. The area of the triangle is half the magnitude of the cross product.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the exact range of $f(x) = \frac{x}{x^2 + 1}$ for $x \in \mathbb{R}$.

Let $y = \frac{x}{x^2 + 1}$. Multiply both sides by $x^2 + 1$ (valid since $x^2 + 1 > 0$ for all $x \in \mathbb{R}$): $y(x^2 + 1) = x \Rightarrow yx^2 - x + y = 0.$ If $y = 0$, $-x = 0 \Rightarrow x = 0$, so 0 is in the range. For $yeq0$, this is a quadratic equation in x. For real x to exist, the discriminant Δ must be non-negative: $\Delta = (-1)^2 - 4(y)(y) = 1 - 4y^2 \geq 0.$ Solve the inequality: $4y^2 \leq 1 \Rightarrow y^2 \leq \frac{1}{4} \Rightarrow\ -\frac{1}{2} \leq y \leq \frac{1}{2}.$ Thu.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 3 · Advanced / AP FRQLet $y = f(x)$ be the particular solution to $\frac{dy}{dx} = xy^2$ with $f(1) = -1$. Use Euler's method with step size $\Delta x = 0.5$ starting at $x = 1$ to approximate $f(2)$, and determine whether the approximation is an overestimate or underestimate by computing $\frac{d^2y}{dx^2}$ at $(1,\ -1)$.

Step 1 · Euler Iteration: • $(x_0, y_0) = (1,\ -1)$. Slope $m_0 = (1)(-1)^2 = 1$. $y_1 = -1 + (1)(0.5) = -0.5$. Point $(1.5,\ -0.5)$. • $(x_1, y_1) = (1.5,\ -0.5)$. Slope $m_1 = (1.5)(-0.5)^2 = 1.5(0.25) = 0.375$. $y_2 = -0.5 + (0.375)(0.5) = -0.5 + 0.1875 =$ **-0.3125**. Step 2 · Second Derivative Evaluation: $\frac{d^2y}{dx^2} = \frac{d}{dx}[xy^2] = (1)y^2 + x(2yy') = y^2 + 2xy(xy^2) = y^2 + 2x^2y^3.$ At $(1,\ -1)$: \[\left. \frac{d^2y}{dx^2} \right|_{(1, -1)} = (-1)^2 + 2(1)^2(-1)^3 = 1 + 2(-1) = 1 - 2 =.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · Medium Find the area of the triangle with vertices $A(1, 1, 0)$, $B(3, 0, 2)$, and $C(0, 2, 4)$.

Find vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2,\ -1, 2\rangle$ $\vec{AC} = \langle -1, 1, 4\rangle$ Compute the cross product: $\vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 2,\ -1, 2;\ -1, 1, 4| = \mathbf{i}(-4 - 2) - \mathbf{j}(8 - (-2)) + \mathbf{k}(2 - 1) = \langle -6,\ -10, 1\rangle$ Magnitude of cross product: $|\vec{AB} \times \vec{AC}| = \sqrt{(-6)^2 + (-10)^2 + 1^2} = \sqrt{36 + 100 + 1} = \sqrt{137}$ The triangle area is half the parallelogram area.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2 + y^2}\, dA$ where D is the semi-annular region $\{(x, y) : 1 \leq x^2 + y^2 \leq 4, y \geq 0\}$.

Convert the semi-annulus to polar coordinates: Radius bounds: $1 \leq r^2 \leq 4 \Rightarrow 1 \leq r \leq 2$. Angle bounds (since $y \geq 0$, upper half-plane): $0 \leq \theta \leq \pi$. Area element: $dA = r\, dr\, d\theta$. Set up the integral: $\iint_D e^{x^2 + y^2}\, dA = \int_0^{\pi}\int_1^2 e^{r^2}\, r\, dr\, d\theta = \left(\int_0^{\pi} d\theta\right)\left(\int_1^2 re^{r^2}\, dr\right)$ Substitute $u = r^2 \Rightarrow du = 2r\, dr$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi}\int_y^{\pi}\frac{\sin x}{x}\,dx\,dy$

The domain is $0 \le y \le \pi$ and $y \le x \le \pi$. This is the triangular region bounded by $y = 0$, $x = \pi$, and $y = x$. Reversing order to Type I: x ranges from 0 to π, while y ranges from 0 to x:

$\int_0^{\pi}\int_0^{x}\frac{\sin x}{x}\,dy\,dx = \int_0^{\pi}\left[y\frac{\sin x}{x}\right]_{y=0}^{y=x}dx = \int_0^{\pi}x\frac{\sin x}{x}\,dx = \int_0^{\pi}\sin x\,dx = [-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos 0) = -(-1) - (-1) = 1 + 1 = 2$

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.

Take natural logarithms of both sides: $\ln y = \ln(x^{\sin x}) = \sin x \cdot \ln x$. Differentiate implicitly with respect to x using the Product Rule on the right: $\frac{1}{y}\frac{dy}{dx} = (\cos x)(\ln x) + (\sin x)\left(\frac{1}{x}\right)$. Multiply both sides by $y = x^{\sin x}$: $\frac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right)$.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

Level 2 · MediumLet $\frac{dy}{dx} = x + 2y$ with $y(0) = 1$. Use Euler's method with step size $\Delta x = 0.2$ to approximate $y(0.4)$.

Step 1 (from $x = 0$ to 0.2): $x_0 = 0, y_0 = 1$. Slope $m_0 = 0 + 2(1) = 2$. $y_1 = 1 + (2)(0.2) = 1 + 0.4 = 1.4$.
Step 2 (from $x = 0.2$ to 0.4): $x_1 = 0.2, y_1 = 1.4$. Slope $m_1 = 0.2 + 2(1.4) = 0.2 + 2.8 = 3.0$.
$y_2 = 1.4 + (3.0)(0.2) = 1.4 + 0.6 = \mathbf{2.0}$. Approximate value $y(0.4) \approx \mathbf{2.0}$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization**

Level 1 · Foundations Evaluate $\iint_R (2x + y)\,dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.

Set up iterated integral: $\int_1^2\int_0^3 (2x + y)\,dx\,dy = \int_1^2 \left[x^2 + xy\right]_0^3 dy = \int_1^2 (9 + 3y)\,dy$ Integrate with respect to y:
$\left[9y + \frac{3y^2}{2}\right]_1^2 = \left(18 + \frac{12}{2}\right) - \left(9 + \frac{3}{2}\right) = 24 - 10.5 = 13.5 = \frac{27}{2}$

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Level 2 · MediumA particle moves with position vector $\vec{r}(t) = \langle t^2 - 4t, \frac{1}{3}t^3 - 4t\rangle$. At what time $t > 0$ is the particle at rest?

A particle is at rest if and only if its velocity vector is $\vec{0}$ (both components are simultaneously 0). • $x'(t) = 2t - 4 = 0 \Rightarrow t = 2$. • $y'(t) = t^2 - 4 = 0 \Rightarrow t = \pm 2$. Since $t > 0$, both components are zero simultaneously at $\mathbf{t = 2}$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Level 3 · Advanced / AP FRQA particle moves along a planar path with velocity $\vec{v}(t) = \langle \sin(t^2), \cos(t^3) \rangle$ for $0 \le t \le 3$. Write the definite integral representing the total distance traveled by the particle, and calculate the numerical value using three decimal places.

Written Setup: Total Distance $= \int_0^3 \sqrt{\sin^2(t^2) + \cos^2(t^3)}\, dt$. Numerical Evaluation: Entering this into a graphing calculator yields: Total Distance $\approx$ **2.874** (or 2.873).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the slope of the tangent line to $x^2y + y^2 = 5$ at the point $(2, 1)$.

a. $-\frac{2}{3}$ ✓ **b.** $-\frac{4}{5}$

c. $-\frac{4}{6}$ **d.** -1

$2xy + x^2y' + 2yy' = 0$. At $(2, 1)$: $2(2)(1) + 4y' + 2(1)y' = 0 \Rightarrow 4 + 6y' = 0 \Rightarrow y' = -\frac{4}{6} = -\frac{2}{3}$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For $x^2 + y^2 = 16$, what is $\frac{d^2y}{dx^2}$ expressed solely in terms of y?

a. $-\frac{16}{y^3}$ ✓ **b.** $\frac{16}{y^3}$

c. $-\frac{1}{y^2}$ **d.** $-\frac{4}{y^3}$

$y' = -\frac{x}{y}$. Differentiate: $y'' = -\frac{1(y) - xy'}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y^2 + x^2}{y^3}$. Since $x^2 + y^2 = 16$, $y'' = -\frac{16}{y^3}$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

At the point $(0, 2)$ on the ellipse $4x^2 + y^2 = 4$, what is $\frac{d^2y}{dx^2}$?

a. -2 ✓ **b.** 2

c. -4 **d.** 0

$8x + 2yy' = 0 \Rightarrow y' = -\frac{4x}{y}$. At $(0, 2)$, $y' = 0$. $y'' = -\frac{4y - 4xy'}{y^2} = -\frac{4(2) - 0}{4} = -\frac{8}{4} = -2$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the sign of the concavity at $(3, 4)$ on the circle $x^2 + y^2 = 25$?

a. **Negative (concave down)** ✓

b. Positive (concave up)

c. Zero (inflection point)

d. Undefined

$y'' = -\frac{25}{y^3}$. At $(3, 4)$, $y'' = -\frac{25}{64} < 0$, which is concave down.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If g is the inverse of f, which formula gives $g'(a)$ where $f(b) = a$?

a. $g'(a) = \frac{1}{f'(b)}$ ✓

b. $g'(a) = f'(b)$

c. $g'(a) = \frac{1}{f'(a)}$

d. $g'(a) = -\frac{1}{f'(b)}$

By the Inverse Function Theorem, $g'(a) = \frac{1}{f'(f^{-1}(a))} = \frac{1}{f'(b)}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = x^3 + 2x - 1$. If g is the inverse of f, find $g'(2)$.

a. $\frac{1}{5}$ ✓

b. 5

c. $\frac{1}{14}$

d. 14

Find x where $f(x) = 2$: $x^3 + 2x - 1 = 2 \Rightarrow x^3 + 2x - 3 = 0 \Rightarrow x = 1$. Then $f'(x) = 3x^2 + 2 \Rightarrow f'(1) = 5$. Thus $g'(2) = \frac{1}{f'(1)} = \frac{1}{5}$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Let $f(x) = x^5 + x + 1$. What is the derivative of f^{-1} at $x = 3$?

a. $\frac{1}{6}$ ✓

b. 6

c. $\frac{1}{80}$

d. $\frac{1}{5}$

Notice $f(1) = 1 + 1 + 1 = 3$. $f'(x) = 5x^4 + 1 \Rightarrow f'(1) = 6$. Therefore, $(f^{-1})'(3) = \frac{1}{6}$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the slope of the tangent line to $y = \arctan(x)$ at $x = 1$?

a. $\frac{1}{2}$ ✓

b. 1

c. $\frac{\pi}{4}$

d. $\frac{1}{\sqrt{2}}$

$y' = \frac{1}{1+x^2}$. At $x = 1$: $y'(1) = \frac{1}{1+1^2} = \frac{1}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $\frac{dy}{dx}$ by implicit differentiation if $2x^2 + 3y^2 = 1$.

a. $\frac{2x}{3y}$

b. $-\frac{2x}{3y}$ ✓

c. $-\frac{3y}{2x}$

d. $-\frac{2x}{3}$

Differentiate both sides: $4x + 6y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{2x}{3y}$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Implicit differentiation of $x^2 + y^2 = 25$ gives $\frac{dy}{dx} =$

a. $\frac{x}{y}$

b. $-\frac{x}{y}$ ✓

c. $\frac{y}{x}$

d. $-\frac{y}{x}$

Differentiate both sides: $2x + 2y\frac{dy}{dx} = 0$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Identify the general solution to the first-order differential equation $\frac{dy}{dx} = 3y$.

a. $y = C\ln(3x)$

b. $y = \frac{3}{2}x^2 + C$

c. $y = Ce^{-3x}$

d. $y = Ce^{3x}$ ✓

Separation of variables $\frac{dy}{y} = 3dx \Rightarrow \ln|y| = 3x + c_1 \Rightarrow y = Ce^{3x}$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the average value of $f(x) = 4x^2$ on $[3, 7]$.

a. $\frac{1264}{3}$

b. $\frac{310}{3}$

c. $\frac{322}{3}$

d. $\frac{316}{3}$ ✓

Average value $= \frac{1}{7-3}\int_3^7 4x^2\,dx = \frac{\frac{1264}{3}}{4} = \frac{316}{3}$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A population grows according to the logistic model $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{315}\right)$. Find the population P at the point of maximum growth.

a. 630

b. 78.75

c. 315

d. **157.5** ✓

Logistic growth is maximal at the point of inflection, where $P = M/2 = 315/2 = 157.5$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $f(x) = \{1x + 3, x < 5; 3x - 6, x \geq 5$, find $\lim_{x \to 5} f(x)$, if it exists.

a. **does not exist** ✓

b. 8

c. 18

d. 9

Left limit $= 8$, right limit $= 9$; since they differ, the limit does not exist.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Classify the discontinuity of $f(x) = \frac{(x-2)(x-4)}{x-2}$ at $x = 2$.

a. infinite (a vertical asymptote)

b. **removable (a hole)** ✓

c. the function is continuous there

d. jump discontinuity

The factor $(x - 2)$ cancels, leaving a removable discontinuity (hole) at $x = 2$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $G(x) = \int_5^{x^3} \sqrt{t+2}\, dt$. By the Fundamental Theorem of Calculus, what is $G'(x)$?

a. $3x^2\sqrt{x+2}$

b. $\sqrt{x^3+2}$

c. $3x^2\sqrt{x^3+2}$ ✓

d. $x^3\sqrt{x^3+2}$

Using FTC Part 1 and Chain Rule, $G'(x) = \sqrt{x^3+2} \cdot (3x^2)$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For the differential equation $\frac{dy}{dx} = 1x - 5y$, find the slope of the solution curve at the point $(3, 2)$.

a. -9

b. -5

c. 7

d. **-7** ✓

Substitute $x = 3$, $y = 2$: $\frac{dy}{dx} = 1(3) - 5(2) = -7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

If $y = 6e^{1t}$ is a solution of $\frac{dy}{dt} = ky$, what is the value of k?

a. 2

b. **1** ✓

c. 3

d. -1

$\frac{dy}{dt} = 1(6)e^{1t} = 1y$, so $k = 1$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:

a. -1

b. -2 ✓

c. 0

d. 1

Differentiating $x^2 + y^2 = 2$ gives $y' = -\frac{x}{y}$ and $y'' = -\frac{y - xy'}{y^2} = -\frac{x^2 + y^2}{y^3}$. At $(1, 1)$, $y'' = -\frac{2}{1^3} = -2$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:

a. **Lagrange multiplier** ✓

b. eigenvalue

c. gradient

d. curvature

Applying Application of Derivatives principles, λ = Lagrange multiplier. Therefore, the verified correct option is Lagrange multiplier.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The magnitude of cross product $|\mathbf{a} \times \mathbf{b}|$ equals the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. For $|a| = 3, |b| = 4, \theta = \pi/6$:

a. 24

b. 12

c. 3

d. 6 ✓

Applying Vectors and Vector-Valued Functions principles, $|a||b|\sin(\pi/6) = 12 \cdot 1/2 = 6$. Therefore, the verified correct option is 6.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Area Between Curves

Cross product $\mathbf{a} \times \mathbf{b}$ is orthogonal to:

a. $\mathbf{a}$ only

b. $\mathbf{b}$ only

c. neither

d. **both $\mathbf{a}$ and $\mathbf{b}$** ✓

$\mathbf{a} \times \mathbf{b} \perp \mathbf{a}$ and $\perp \mathbf{b}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Implicit differentiation of $xy = 5$: $\frac{dy}{dx} =$

a. $\frac{5}{x}$

b. $\frac{y}{x}$

c. $-\frac{y}{x}$ ✓

d. $\frac{x}{y}$

$y + x\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -y/x$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Limits Graphically and Numerically

Integration of inverse functions (Laisant-Hermite): $\int_a^b f^{-1}(y)dy = bf^{-1}(b) - af^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f(x)dx$.

Example for $f = e^x$:

a. gives nothing useful

b. yields: $\ln b - \ln a$

c. gives $e^b - e^a$

d. **yields:** $b\ln b - a\ln a - (b - a)$ ✓

Inverse function integration applied to $\ln$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

The cross product of two parallel vectors is:

a. A unit vector

b. **The zero vector** ✓

c. Their dot product

d. Undefined

Parallel vectors give $\vec{0}$ cross product.

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